Exact Necklace Periods and a Koopman Owner for a Margolus Automaton

Route-A structural certificate C165

Abstract

We study binary configurations on a ring of $2m$ sites under two staggered layers of nearest-neighbour swaps. One full tick first exchanges the even pairs and then the staggered pairs. Direct composition sends even labels two sites forward and odd labels two sites backward. By pairing each even site with a reversed odd site, we identify the configuration map with cyclic rotation of an m-letter word over a four-letter alphabet. It follows that $|\operatorname{Fix}(T^n)| = 4^{\gcd(m,n)}$, and Möbius inversion gives all exact-period configurations and primitive cycles. These are all-size source theorems, not extrapolations from the finite checks. Proper periods occupy at most $m/2^m$ of configuration space, uniformly including the identity boundary $m = 1$. Reflection reverses the full tick, while the finite counting-measure Koopman permutation owns the inverse source zeta and admits an antiunitary reversal. This is not a chaos, target-divisor, or uniform self-adjoint spectral claim.

Keywords: partitioned cellular automaton; Margolus blocks; cyclic rotation; necklace; exact period; primitive cycle; dynamical zeta.

中文摘要

本文研究 $2m$ 个站点的二元环，其中一个完整时钟依次执行两层交错近邻交换。 直接跟踪带标签的元胞可知，偶站点向前移动两格，奇站点向后移动两格。 把偶站点与反向编号的奇站点配对后，整个配置动力系统严格等价于四字母长度 m 词的循环旋转。因此可以得到所有时刻的不动配置数，并由莫比乌斯反演恢复 精确周期配置和本原周期。有限枚举仅用于回归检查，不承担全参数定理的证明。 所有非满周期配置的比例至多为 $m/2^m$，并且该估计保留 $m = 1$ 的恒等边界。 站点反射把完整时钟变为其逆，而计数测度上的有限维Koopman 置换恰好拥有源动力行列式及反酉时间反演。本文不宣称混沌，不比较目标除子， 也不把该有限酉算子提升为自伴目标算子。

关键词：分块元胞自动机；交错交换；循环旋转；项链；精确周期； 本原周期；动力 ζ 函数。

1 Full-tick site law

Let $X_m = \{0,1\}^{\mathbb{Z}/(2m)\mathbb{Z}}$. Layer A swaps $(0,1),(2,3),\ldots$, while B swaps $(1,2),(3,4),\ldots,(2m-1,0)$. One tick is $T = B \circ A$. The labelled-cell permutation is

$$\tau(i) = \begin{cases} i+2, & i \text{ even}, \\ i-2, & i \text{ odd}, \end{cases} \pmod{2m}. \tag{1}$$

Put $e_j = 2j$ and $o_j = 1-2j$. Then $\tau(e_j) = e_{j+1}$ and $\tau(o_j) = o_{j+1}$. Hence

$$\Phi(x)_j = (x_{e_j}, x_{o_j}) \in \{0,1\}^2 \tag{2}$$

conjugates T to cyclic rotation of a four-letter word of length m.

2 Period law

The nth power of the rotation has $\gcd(m,n)$ coordinate cycles, proving

$$|\operatorname{Fix}(T^n)| = 4^{\gcd(m,n)}. \tag{3}$$

Every exact period divides m. For $d \mid m$,

$$P_m(d) = \sum_{e \mid d} \mu(d/e) 4^e, \qquad C_m(d) = P_m(d)/d. \tag{4}$$

Thus the finite source zeta is

$$\zeta_T(z) = \prod_{d \mid m} (1 - z^d)^{-C_m(d)}. \tag{5}$$

At $m = 1$ the full tick is the identity on four configurations. At $m = 2$ there are four fixed configurations and twelve configurations in six two-cycles.

3 Uniform full-period concentration

If $d < m$ divides m, then $d \le m/2$ and $P_m(d) \le 4^d$. There are fewer than m proper divisors, so

$$\Pr(\text{per} < m) = \frac{\sum_{d|m,\ d<m} P_m(d)}{4^m} \le m4^{-m/2} = \frac{m}{2^m}. \tag{6}$$

Thus $\Pr(\text{per} = m) \ge 1 - m/2^m$. At $m = 1$ the short set is empty. The factor m is a deliberately coarse uniform majorant; no optimality is claimed.

4 Reversal and the same-clock operator owner

Reflection $r(i) = -i \pmod{2m}$ preserves parity, and (1) gives $r\tau r = \tau^{-1}$. Its configuration action R obeys $RTR = T^{-1}$. On $\mathcal{H}_m = \ell^2(X_m)$ with counting measure, let $(U_T f)(x) = f(T^{-1}x)$. Each dynamical d-cycle contributes a cyclic permutation block, hence

$$\det(I - zU_T) = \prod_{d|m} (1 - z^d)^{C_m(d)} = \zeta_T(z)^{-1}. \tag{7}$$

For $\Theta f(x) = \overline{f(Rx)}$, one has $\Theta U_T \Theta = U_T^{-1}$. This is a finite, source-natural, same-clock Koopman lift. It is self-adjoint for $m = 1, 2$: respectively $T = I$ and $T^2 = I$. For every $m \ge 3$, $P_m(m) > 0$ supplies an m-cycle, so $T^2 \ne I$ and the permutation unitary is not self-adjoint. No uniform self-adjoint Hilbert–Pólya realization is claimed.

Exact ledgers contain 136 fixed-time and 50 period cells through $m = 16$; 87,380 configurations are directly enumerated through $m = 8$. The independent checker passes 723 assertions, SymPy passes 481 checks, byte replay is exact, and 58 hostile cases are rejected. These are regression sentinels. The strict tuple is (`A1_WEAK`, `A2_FAIL`, `A3_FAIL`, `A4_NATURAL_QUANTIZATION`). The exact conjugacy is not described as chaos or interaction. No target table, Euler factor, root number, automorphy, Hilbert–Pólya operator, or Route-B input is used. Scope:

`NO_BAD_EULER_OR_ROOT_NUMBER`.

Declarations

Data and code availability. Package-local exact evidence and code accompany the manuscript.

Ethics. No human, animal, clinical, personal, or sensitive data are used; approval is not applicable.

Author contributions (CRediT). This anonymous certificate records Conceptualization, Formal analysis, Software, Validation, Writing—original draft, and Writing—review and editing. AI systems are not authors.

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